

Slide 1 — Scanned, Guide-Signed Title Slide (Mandatory)

- Before the Review-I, obtain the guide's signature with date on the title slide (next slide).
- Scan or photograph the signed title slide clearly.
- Replace this slide with the scanned copy, so that it remains Slide 1 of the presentation.
- This requirement applies to every review of Project-I (2026).

GaN-Based Power Converter

A segmented gate driver for the GaN half-bridge inside it

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Programme / School:

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Review & Date:

Review-I · 09.09.2026 (Wednesday)

Guide's Signature with Date

Signature: _____

Date: _____

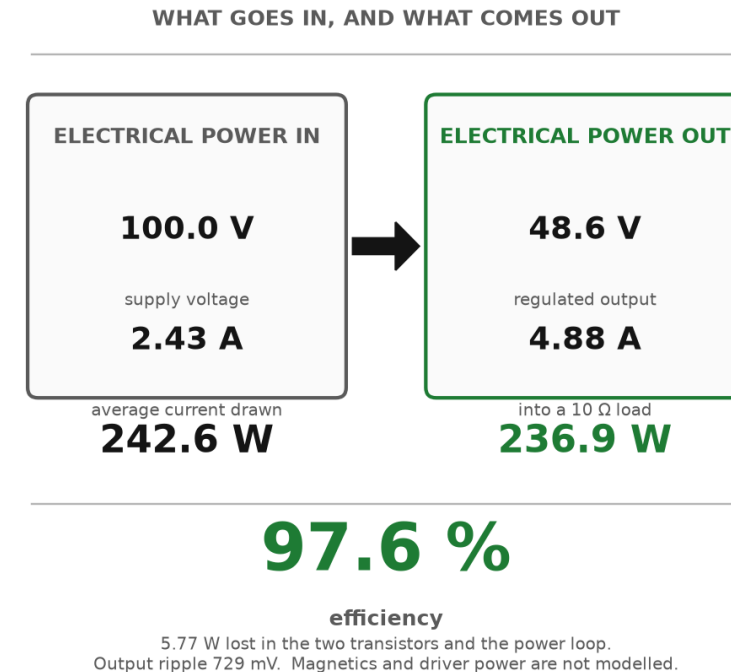
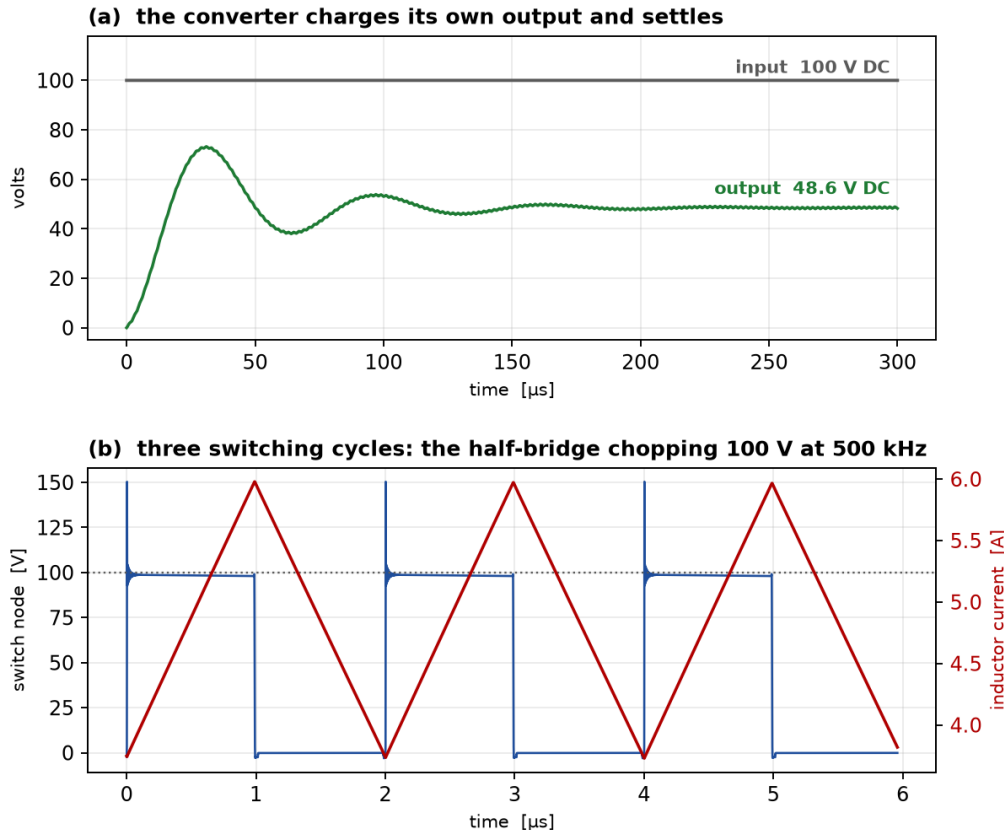
Approval is mandatory before every review. After signing, scan this slide and keep the scanned copy as Slide 1 of the presentation.

Review-I — 09.09.2026 (5 Marks, 5%)

- Focus: Proposed Solution & 50% Work Completion.
- Duration: 10 minutes.
- Presentation contents:
 - Signed & scanned title slide (Slide 1)
 - Problem statement and background
 - Literature survey — minimum 8–10 recent references
 - Existing solutions and their limitations
 - Proposed solution, scope and methodology
 - 50% work completed, timeline, tools & technologies
- Mark split-up: Problem clarity & relevance (2), Literature survey quality (1), Proposed solution feasibility (1), Presentation quality (1).

What we are building — the converter

GaN synchronous buck converter — `sim/buck.cir`, `ngspice` • 100 V DC → 48.6 V DC at 4.88 A



A GaN synchronous buck converter. 100 V DC goes in; the two GaN transistors chop it at 500 kHz; the filter turns that back into DC. 48.6 V at 4.88 A comes out — 236.9 W into the load from 242.6 W drawn, so 97.6 % of the power gets through. 5.8 W is lost in the transistors and the power loop.

Open loop, no feedback controller yet — the overshoot on the way up in panel (a) is that. Reproduce: `ngspice -b sim/buck.cir`, then `python3 scripts/buck_figure.py`

01

The problem

A real failure, and the question nobody has answered

Problem Statement & Background

- **Problem Statement:**

- When one GaN transistor switches, its fast voltage swing pushes charge into the gate of the other transistor — the one that is supposed to stay OFF. That gate rises to **1.65 V**, and **only 1.4 V is needed to turn it on**. Both devices conduct at once and the supply shorts through them.

- **Background & Significance:**

- GaN half-bridges sit inside EV inverters and battery chargers. GaN is chosen because it switches fast — and that speed is exactly what causes this fault.

- **Existing Solutions:**

- Gate drivers that turn the device on in steps, clamp the off gate down, adjust the gap between the two switching edges, and hold the gate at -2 V [1]–[4]. Reported: 30.5 % less overshoot, 75 % less turn-off loss [5]–[7].

- **Limitations / Research Gap:**

- **Nobody has measured what the re-tuning is worth.** Those gains mix two different things: picking one good setting, and changing the setting while the converter runs. Only the second needs a sensor, a lookup table and a controller.

The base paper we build on

Takayama, Okuda & Hikihara (2022)	
What it does	Drives the gate from a multibit CODE instead of a resistor — the driver is a DAC, and the code changes DURING the switching edge.
Why it matters to us	A digital code is finite. That is what makes the setting space searchable, and what makes the driver implementable on an FPGA rather than as an analogue network.
What it leaves open	It never asks whether the code has to CHANGE as load, voltage and temperature move. That question is this project.

H. Takayama, T. Okuda and T. Hikihara, “**Digital Active Gate Drive of SiC MOSFETs for Controlling Switching Behavior,**” Int. J. Circuit Theory Appl., vol. 50, no. 1, pp. 183–196, 2022. doi.org/10.1002/cta.3136

The Five Closest — and the base paper we replicate

Ref	Author(s) & Year	Title / Source	Aim — what the paper sets out to do	Gap we found → what WE fill
BASE	H. Takayama, T. Okuda & T. Hikiyara (2022)	Digital Active Gate Drive of SiC MOSFETs for Controlling Switching Behavior — Preparation Toward Universal Digitization of Power Switching Int. J. Circuit Theory Appl. , vol. 50, no. 1, pp. 183–196 doi.org/10.1002/cta.3136	A DAC-inspired driver: a multibit gate signal SEQUENCE sets the gate waveform digitally, so switching behaviour is chosen by a code rather than by a resistor.	THE BASE PAPER. Its multibit gate code is the direct ancestor of our 720-point control word, and it is digital — implementable on an FPGA — rather than a fixed analogue network. We replicate its premise on GaN and then ask the question it does not: of the gain a code buys, how much needs per-operating-point adaptation at all? WE FILL: the exhaustive 720-word search that makes the fixed-vs-adaptive split measurable, and the GaN third-quadrant cost their SiC device never pays.
10	Zhang, Yu, Leng, Cui, Deng & Ng (2020)	A Segmented Gate Driver for E-mode GaN HEMTs with Simple Driving Strength Pattern Control Proc. IEEE ISPSD, 2020 , pp. 102–105 ieeexplore.ieee.org/document/9170108	Segmented output stage on E-mode GaN: 7 slices, pattern timing 0.5–5 ns, strength set by one external bias resistor	Architecturally the closest published driver to ours — segmented slices, pattern-selected strength. It is an ASIC with a fixed pattern set; our contribution is to search the whole pattern space exhaustively and price what the search actually buys. WE FILL: the price of the pattern space: what an exhaustive search buys over their fixed pattern set.
11	Wang, Tao, Xiao, Luo, He, Zhou, Zhang & Wang (2024)	High-Frequency Three-Level Gate Driver for GaN HEMT Bridge Crosstalk Suppression IEEE Trans. Power Electron. , 39(1), 1343–1352 ieeexplore.ieee.org/document/10286072	Three-level drive; capacitor–diode negative rail with a digitally-clamped zero level, to 5 MHz	Recent, GaN, and on exactly our failure mode. It suppresses crosstalk with added passives; we get the same protection from the Miller clamp already in the cell and measure its price at 0.04 %. WE FILL: whether the setting has to change with the operating point at all. It does not: 3.9 % of baseline.
12	Wang, Sun, Yan, Ma & Xu (2024)	An Integrated Suppression Method of Both Gate-Source Voltage Oscillation and Crosstalk for GaN HEMT Gate Driver IEEE Trans. Power Electron. , 39(11), 14643–14655 ieeexplore.ieee.org/document/10591431	One driver addressing gate-source ringing and crosstalk together	Confirms the two effects are coupled, which is why our cost function prices loss and overshoot jointly rather than optimising either alone. WE FILL: both effects in ONE cost function, showing the objectives genuinely conflict (Pareto, slide 16).
13	Cai, Ye, Lv & Chen (2026)	A High-Efficient GaN Driver With Hybrid Adaptive Dead-Time Control and Peak Delay Control for Synchronous Buck Converter IEEE Trans. Power Electron. , 41(1), 279–290 ieeexplore.ieee.org/document/11146698	Hybrid adaptive dead-time plus peak delay control on a GaN synchronous buck	The most recent adaptive dead-time driver we found, and it adapts the one field our leave-one-out test shows carries the whole benefit — and only at light load. WE FILL: which field is worth scheduling. Only dead time, and only at light load; drive strength is worth 0.00 %.

[9] is the BASE PAPER: we reproduce its premise — a gate waveform chosen by a multibit code — on GaN, then extend it. All five are verified against the publisher record.

The gap this project fills

Every paper on these drivers reports ONE number: how much better it is than a plain driver.
That number hides two separate effects, and they cost very different hardware.

Effect 1 — a better FIXED setting

Pick one good setting once, at design time. Costs nothing while running: no sensor, no ADC, no lookup table, no controller.

25.1 %
of baseline

Effect 2 — ADAPTING per operating point

Change the setting as load, voltage and temperature move. This is the part that needs the sensing hardware.

3.9 %
of baseline

THE GAP: nobody separates them. No paper says how much of the benefit actually needs the re-tuning, because separating the two means running every setting at every operating point — and nobody has done that.

WHY IT MATTERS: only Effect 2 pays for the sensor, ADC and lookup table. If it is small, that hardware is mostly buying something a design-time choice already gives you.

WHAT WE DO: 720 settings × 4 operating points, all of them, in ngspice — then report the two numbers separately instead of adding them together.

02

What we built

A 720-point control word, and an exhaustive search

Proposed Solution & Methodology

- **Aim:**

- To measure how much of a gate driver's benefit comes from **changing its settings while running**, and how much comes from simply **picking one good setting and leaving it**. We test every setting instead of arguing the case.

- **Proposed Solution:**

- A GaN gate driver with 720 possible settings: 8 pull-up steps, 8 pull-down steps, dead time 5–35 ns, a gate clamp, and a –2 V off rail.
- **Method:** run all 720 settings at all four operating points, then compare the best-at-each-point against the best single setting. That difference is what re-tuning is worth, and nothing else.

- **Methodology / Approach:**

- Double-pulse test in ngspice, using a GaN model built from datasheet numbers.
- Eight measurements per run, taken by one script fixed early. Every number must stay the same when the timestep changes by 5× — checked, not assumed.

- **Project Scope:**

- **In scope:** driver design, the setting search, transistor-level output stage in SKY130. **Out of scope:** making the GaN device, sensor hardware, PCB build.

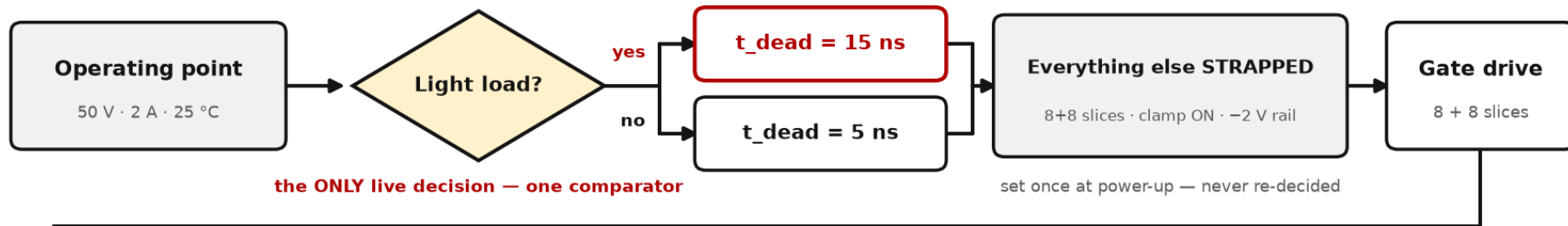
How it works — one use case, start to finish

USE CASE

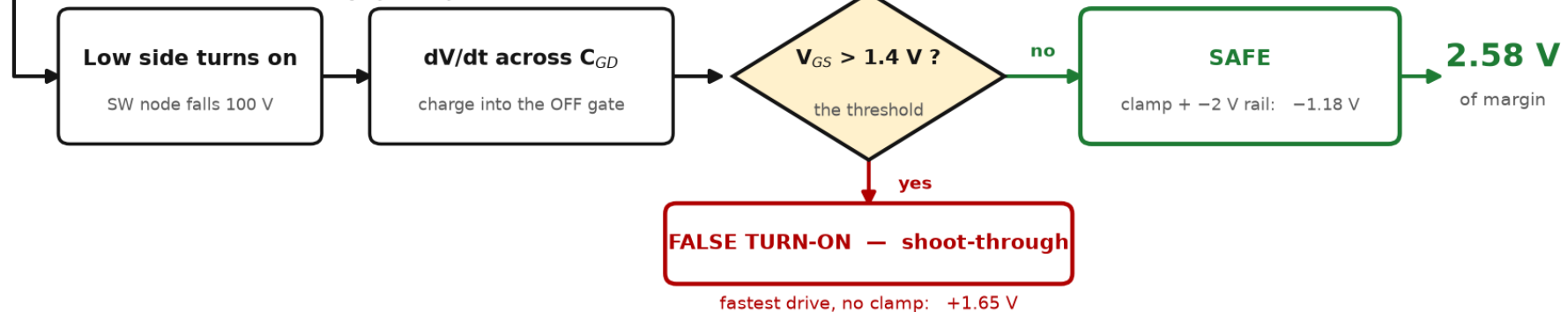
Battery-storage converter — load falling 10 A → 2 A as the pack nears full charge.

Light load is the only one of the four corners where the best dead time differs. This is where the adaptive question is decided.

EVERY SWITCHING EDGE — what the controller decides



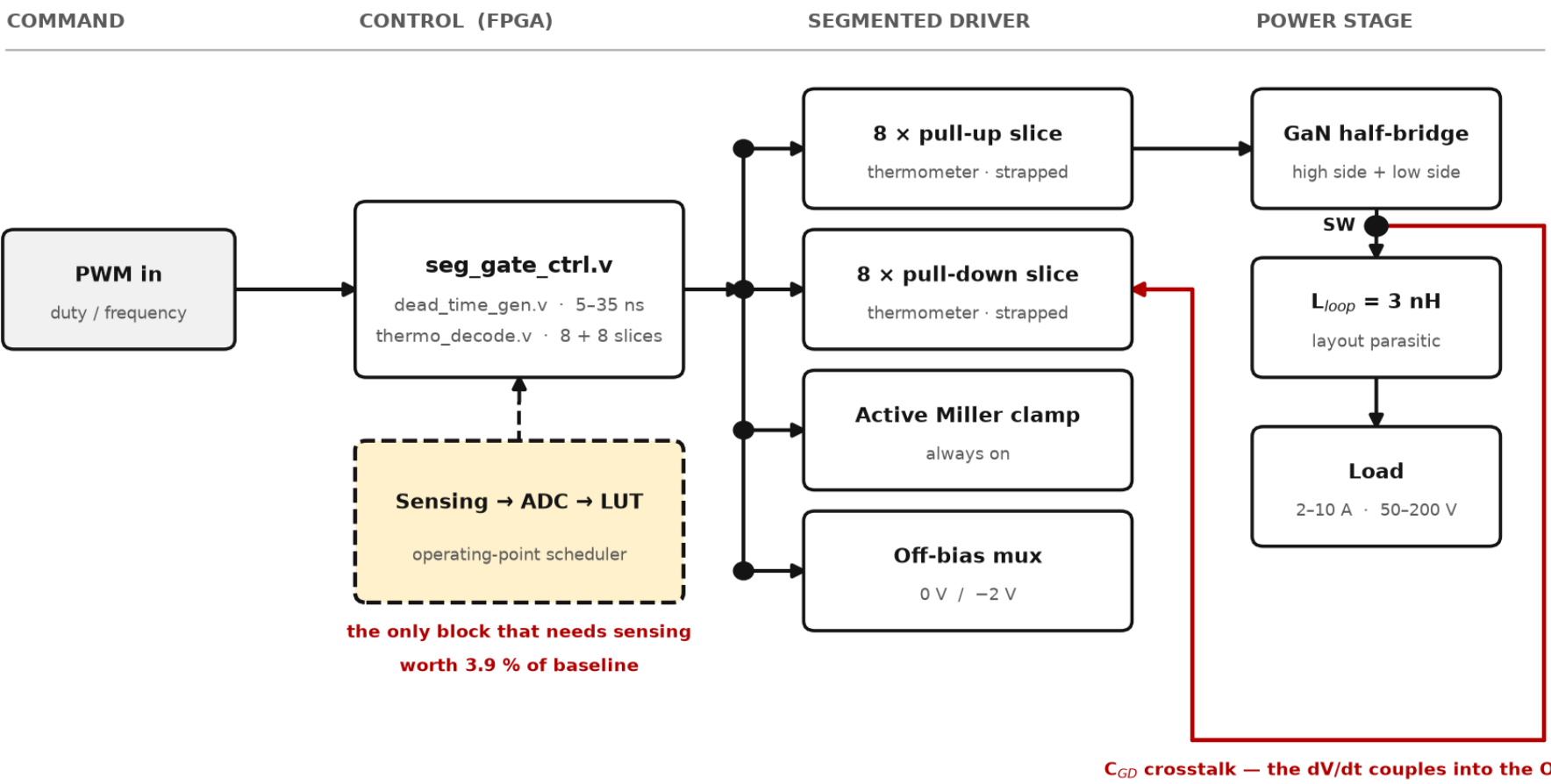
IN THE CIRCUIT — what physically follows



Everything adaptive reduces to the one shaded decision. Freezing it costs 5.45 % across four corners — and 0.00 % if the light-load corner is left out.

Read the top row as the controller and the bottom row as the circuit. The one shaded diamond is the entire adaptive machinery.

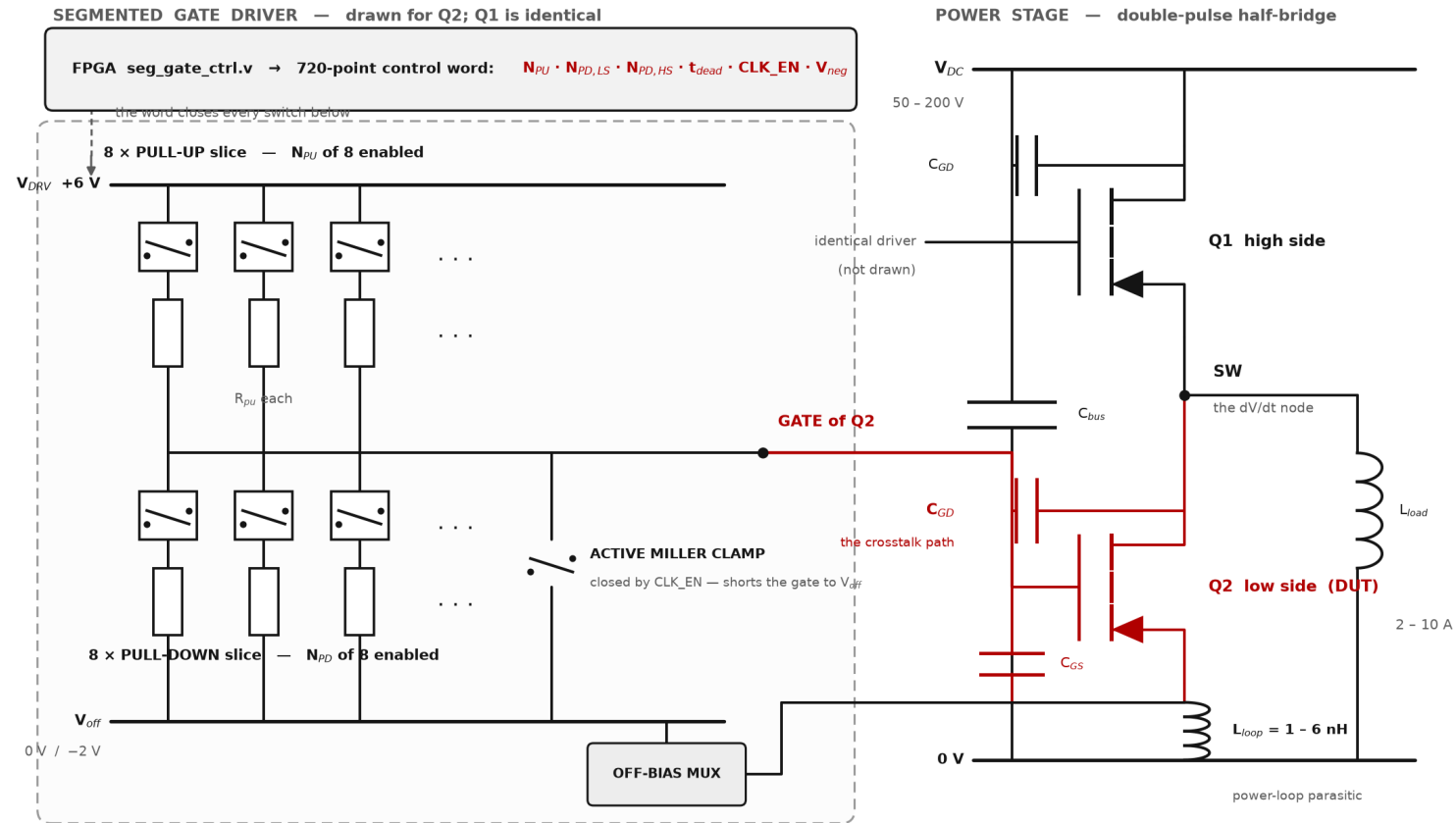
System Architecture



C_{GD} crosstalk — the dV/dt couples into the OFF gate

Solid blocks are set once at power-up — no sensing, no lookup table. The dashed block is the adaptive machinery, and 72 % of what it buys needs only one comparator.

The circuit that is simulated



Every element here is in sim/dpt.cir. Q2 is the device under test; C_{GD} on Q2 is the crosstalk path that turns it on when Q1 switches. GaN has no body diode, so holding V_{off} negative is paid for again across the dead time.

Everything shown is in sim/dpt.cir — that is the file to open in LTspice, and it contains the gate clamp. The three ltspice/*.asc sheets only show how the crosstalk happens and do not draw the clamp. The red CGD on Q2 is the path the charge takes. GaN has no body diode, so the gate has to be held down for the whole dead time.

We implemented the base paper, then ours

Citing a base paper is not a comparison, so we built theirs too. **models/basedrv.lib** is Takayama, Okuda & Hikiyama's driver: a multi-bit code that changes DURING the switching edge, with no gate clamp and no -2 V rail, because those two are ours. It runs inside the same sim/dpt.cir, so only the driver differs.

Base paper, as we built it	+0.534 V	Their changing-in-time code on its own already clears the 1.4 V threshold.
Ours, constant code, no clamp	−0.249 V	FALSE TURN-ON. A fast fixed code is worse than their timed one — their idea is real, and we reproduce it.
Ours, active Miller clamp on	+0.570 V	Only just past the base paper. The clamp on its own is not the story.
Ours, clamp + -2 V off-bias	+2.576 V	4.8× the base paper's margin. This is the version we ship.

What the comparison shows. The base paper shapes the gate over TIME; we keep the code steady and add two things they do not have. Both clear the threshold — theirs works — but ours clears it by 4.8× more, and ours is the one whose settings can then be searched in full.
Reproduce: python3 scripts/basepaper_compare.py — four ngspice runs, prints this table.

Work Completed — 50 %

- **The fault is reproduced, and fixed.**
 - The off gate reaches 1.65 V against a 1.4 V threshold. With the clamp and the –2 V rail: 2.58 V of margin.
- **Every setting has been tested.**
 - 720 settings at each of four operating points — 2,880 runs — so the best at each point is the true best, not the best of a short list.
- **Stress-tested, not claimed.**
 - 21,600 more runs across five device parameters: the answer stays between 4.3 % and 7.7 %. Board inductance and EMI were tested too.
- **Checked in four tools.**
 - LTspice matches ngspice within 2 mV. MATLAB and Octave agree exactly. Vivado: 20 LUTs, 200 MHz met.
 - **In total:** 34,622 simulation runs.

Timeline, Milestones & Tools

- **Where we are**

- **Review-I is done.** The fault is reproduced and fixed, all 720 settings have been tested at every operating point, and the answer has been stress-tested against the device model, the board layout and EMI.

- **What we aim to do next**

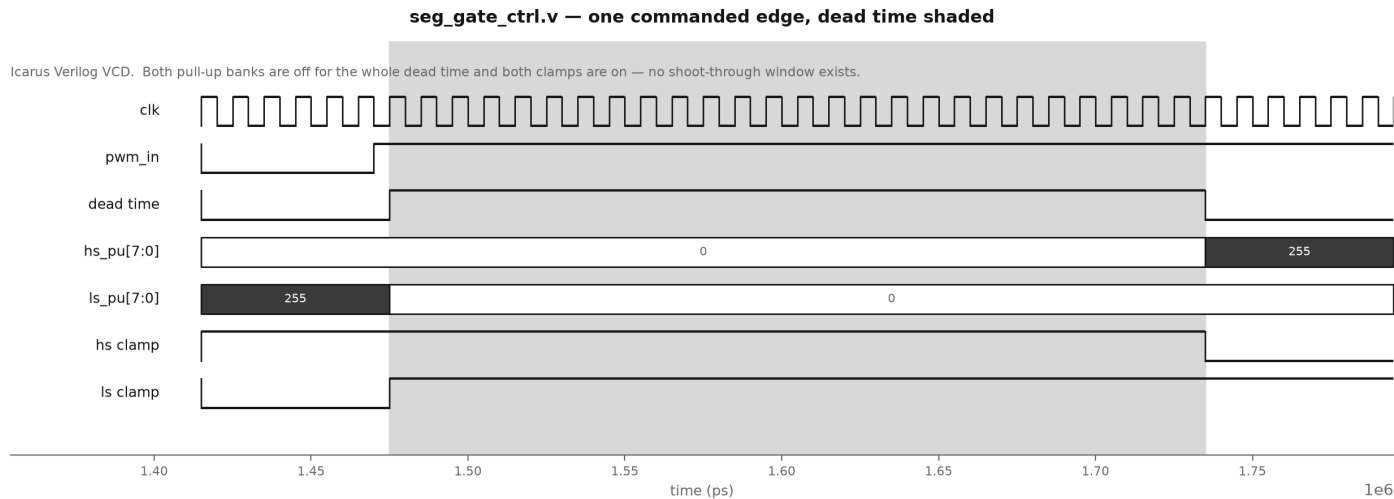
- **Review-II — build it in silicon, not on an FPGA.** A 200 MHz FPGA can only move the dead time in 5 ns steps. Light load wants 15 ns and everything else wants 5 ns, so the steps are too coarse exactly where the benefit is. Drawing the output stage transistor by transistor in SKY130 removes that limit.
- **Review-III — measure one operating point on real hardware.** Build the half-bridge and measure the off-gate voltage against the simulated 1.65 V. Until that exists this is a simulation study, and the title says so.
- **The risk we already know.** A 1.8 V / 3.3 V teaching PDK cannot take a 5 V gate supply. Getting a 5 V-capable PDK is the first milestone, and the one that can hold up the rest.

- **Tools, and how to check any of it**

- ngspice 42 · LTspice 24 · Icarus Verilog · Xilinx Vivado 2024.1.2 · Cadence · SKY130 · Python · MATLAB and Octave. Every number can be regenerated by a named script at github.com/Amritha902/gan-driver.

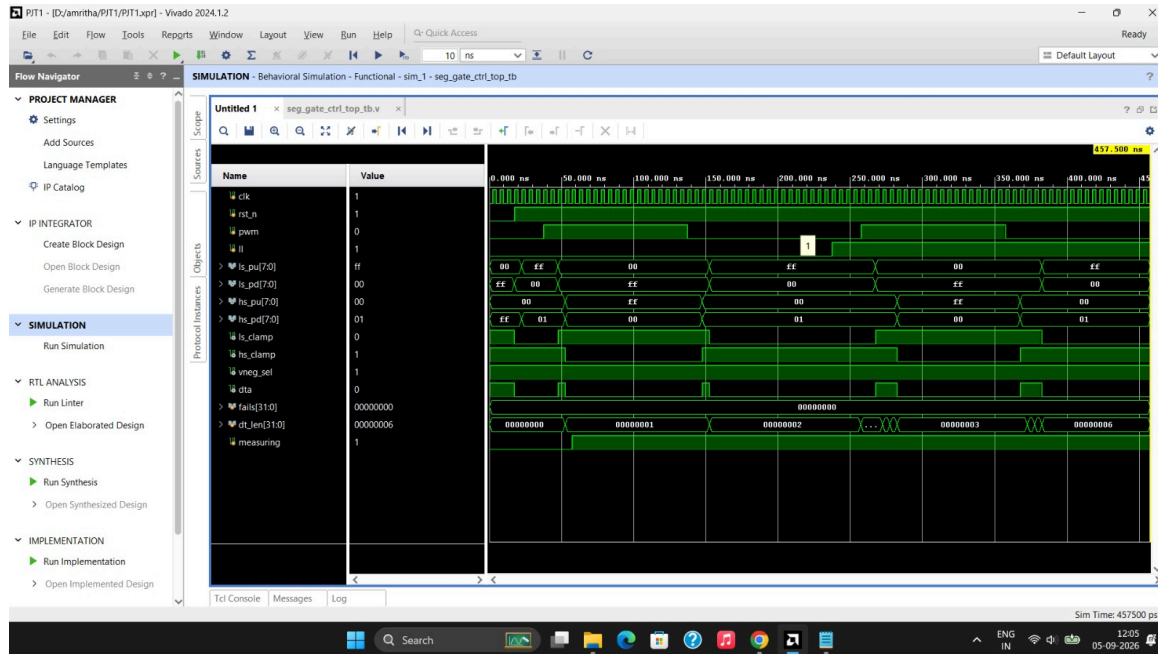
FPGA Controller — verified RTL

- Three Verilog modules that produce exactly the 720 settings the SPICE model uses.
- Dead time is a **live register**; drive strength is **fixed at power-up** — worth 5.45 % against 0.00 %.
- Eight checks pass in Icarus Verilog, safe reset included. A deliberately broken version is caught 221 times (sh rtl/mutate.sh).
- **Built in Vivado 2024.1.2** on an xc7a35t FPGA: **20 LUTs, 20 flip-flops** — 0.10 % of the chip. 200 MHz timing is **met, with 1.996 ns to spare**.
- The 34 failing paths are all chip-output paths against a placeholder 4 ns constraint — the output buffer alone takes 3.49 ns.

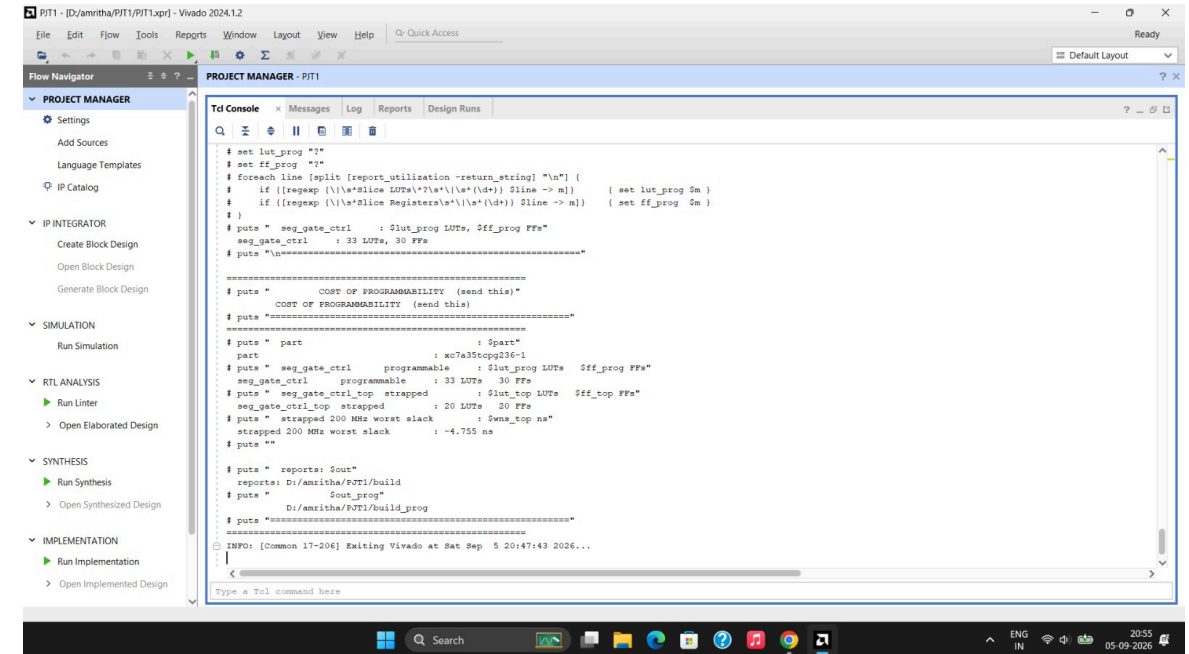


Icarus Verilog VCD. ls_pu falls to 0 before hs_pu rises — the shaded dead time is the gap, and no shoot-through window exists.

Vivado — simulation and synthesis, on screen



Behavioural simulation. fails[31:0] = 0 — every asserted property holds in Vivado's own simulator, a third tool after Icarus and the mutation test.



Synthesis console. 33 LUTs programmable against 20 strapped — the cost of programmability, printed by the tool.

03

What we found

Choosing well beats adapting — and by how much

Result 1 — crosstalk is real; the clamp fixes it

Crosstalk at low-side turn-on, 100 V / 10 A

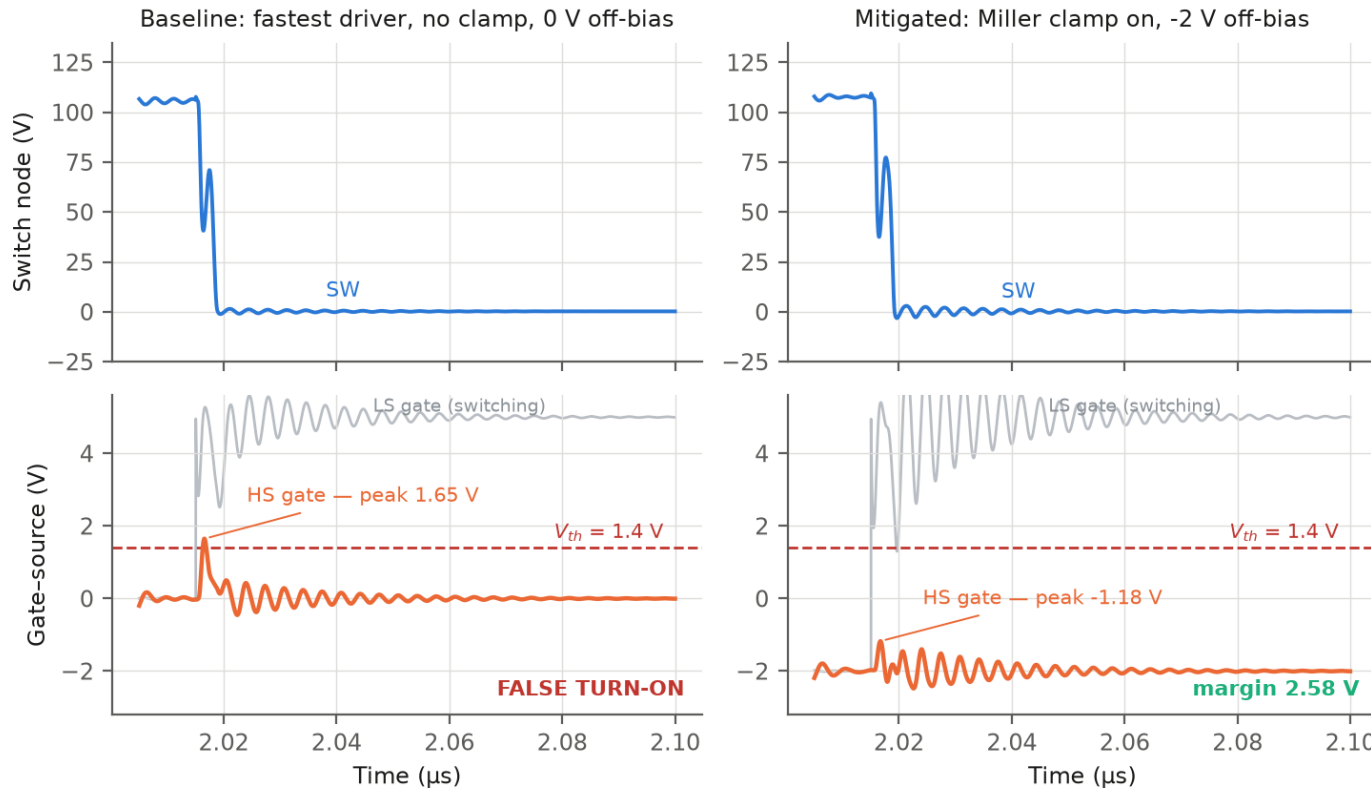


Fig. 1 Voltage on the gate that is supposed to stay OFF: the failure, and the fix that removes it.

The problem is real, and reproduced

1.65 V

peak on the gate that should be OFF

The threshold is 1.4 V — so it turns on by mistake

2.58 V

margin with the clamp and the -2 V rail

The fault is removed completely

And safety is nearly free

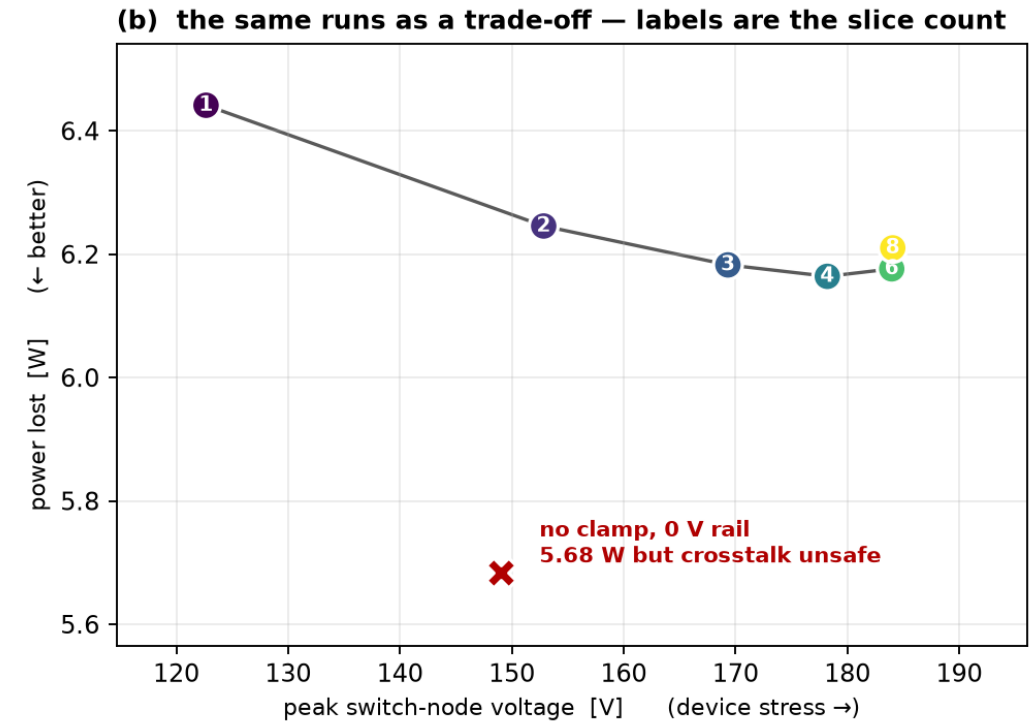
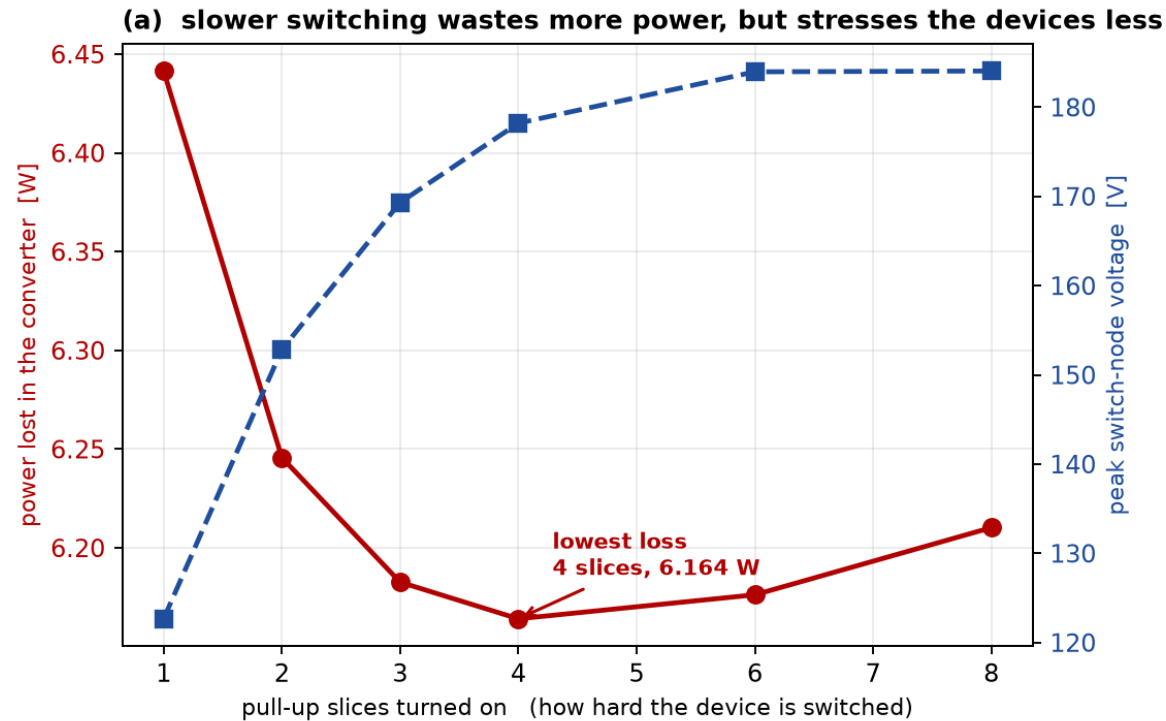
The lowest-energy setting is already safe at three of the four operating points. At the fourth, safety costs 0.04 % more energy — 3.335 μJ against 3.334 μJ.

The clamp is worth 9.7–12.2 % of the blended cost, with no sensor and no controller.

Result 2 — the driver setting moves the converter

The gate-driver setting, measured on the running converter — 8 ngspice runs of sim/buck.cir

Loss moves 4.5 % and peak device voltage moves 50 % across the same knob. Crosstalk safety costs 0.53 W, or 0.22 points of efficiency.



This is why the gate driver is worth studying. The same knob moves power lost by 4.5 % and peak device voltage by 50 %, in opposite directions — there is no setting that is best at both. Lowest loss is 4 slices, not the fastest. Turning the clamp and the -2 V rail on costs 0.53 W, which is 0.22 points of efficiency, and that is the price of crosstalk immunity.

Eight converter runs. Reproduce: `python3 scripts/buck_sweep.py`

Result 3 — re-tuning is worth only 5.2 %

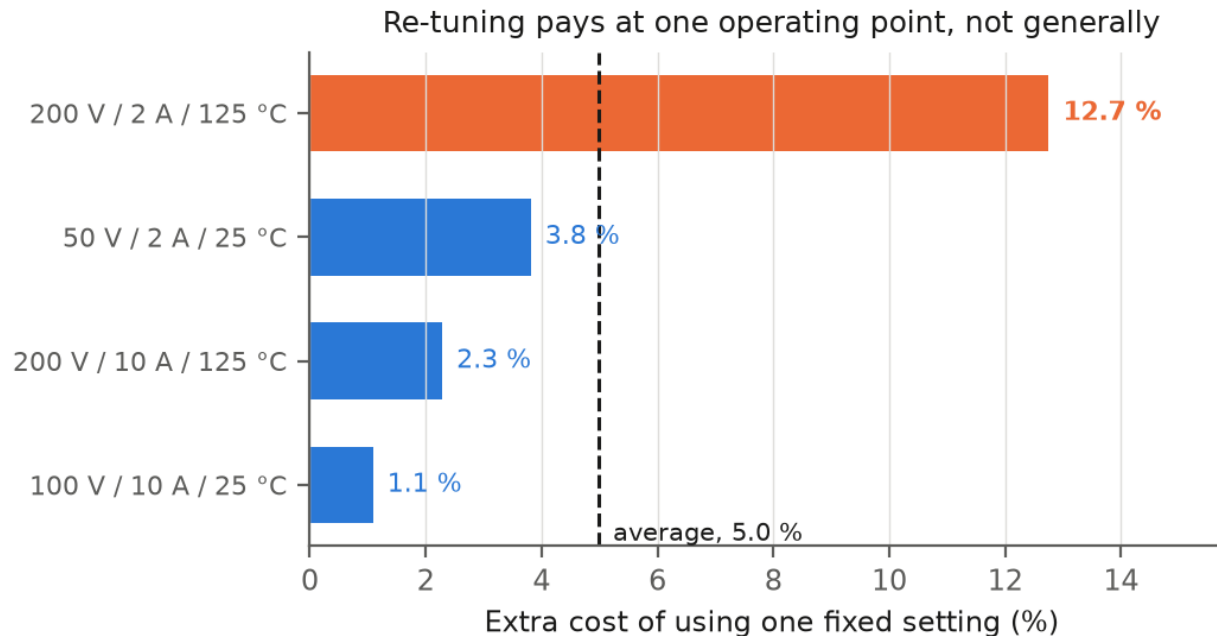


Fig. 2 What it costs to use one fixed setting instead of the best setting at each operating point.

All 720 settings run at every operating point — 2,880 runs — so the best at each point is the true best, not the best of a short list.

What re-tuning is actually worth

5.2 %

the most that re-tuning can ever gain

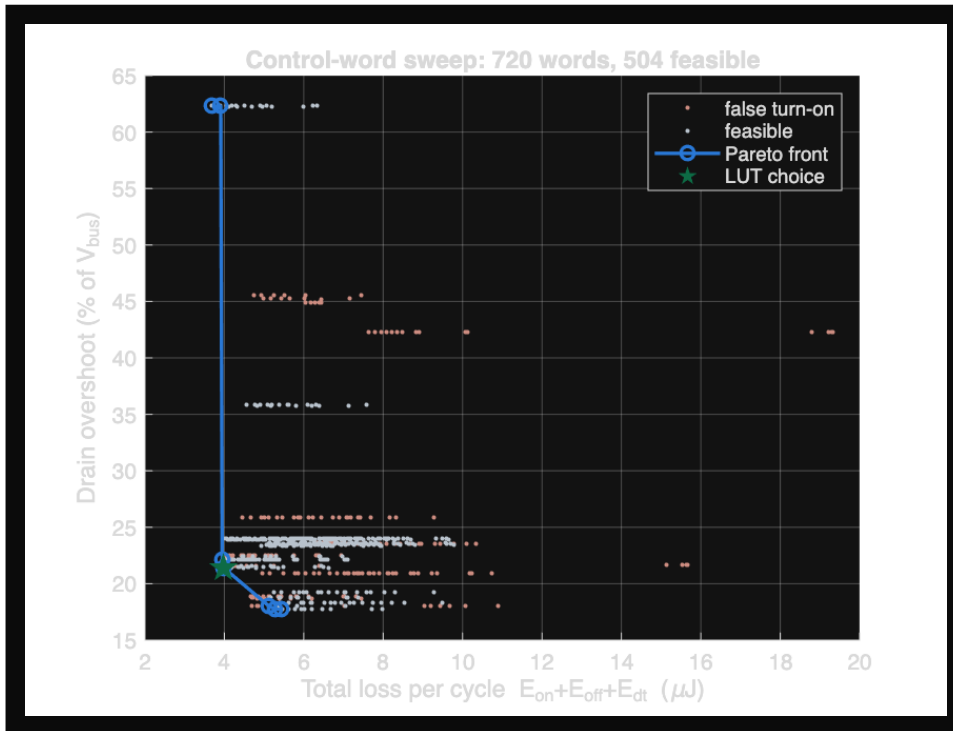
against the best single fixed setting (= 3.9 % of baseline). One fixed setting is nearly as good.

The benefit is not spread out. Three operating points lose only 1–4 % from a fixed setting; one loses 12.7 %.

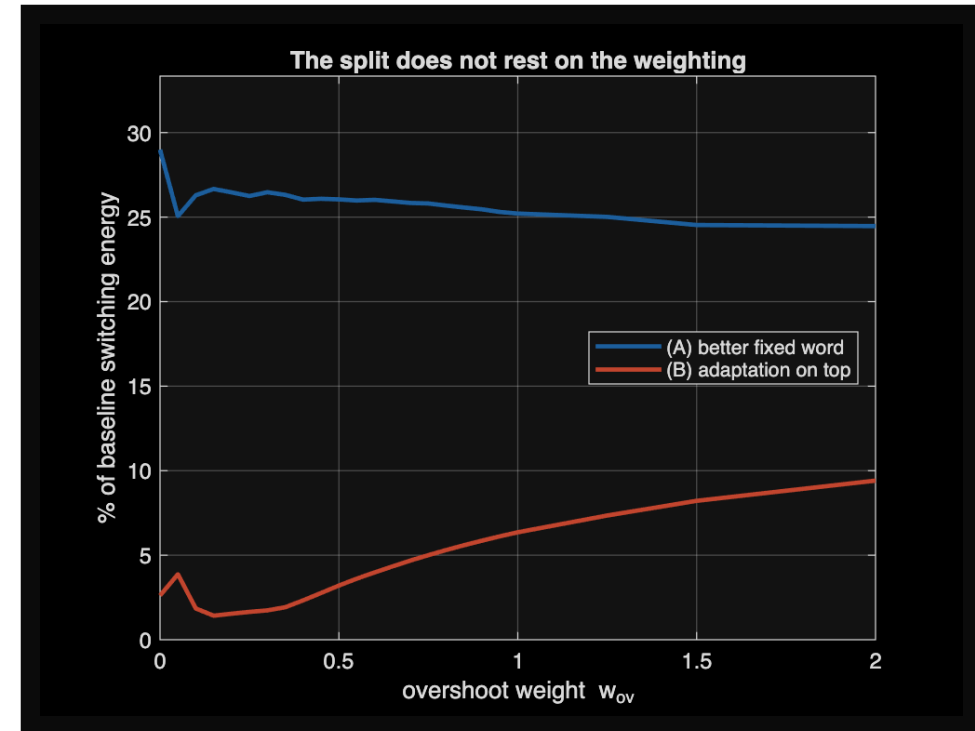
5.2 % is the generous figure — on a finer 36-point grid it is 2.0 %. It all comes from the dead time, and that from one operating point. Freezing the dead time costs 5.45 % across four points. Drop the light-load 50 V / 2 A point and it costs 0.00 %: three points want 5 ns, only that one wants 15 ns. So the adaptive hardware shrinks to one light-load detector choosing between two dead times. Freezing the drive strength costs 0.00 % — and that is what these papers actually adjust.

Steady against the device, not the board. Across 21,600 runs no device parameter moves the answer outside 4.3–7.7 %. Halving the board inductance takes it to 13.5 %, and that is layout, not the transistor [3].

MATLAB — the trade-off curve and the model check



720 settings, 504 of them safe (70 %). The blue line is the best-possible trade-off: buying back one point of overshoot costs 0.039 μJ of energy. The green marker is the setting our cost function picks.



(A) stays above (B) however the two goals are weighted. Picking the setting well is worth 22.5–29.0 % of baseline across the whole range; re-tuning it per operating point 1.3–12.7 %. The order never flips.

Dead time stops helping at about 15 ns: the gain per extra nanosecond runs $269 \rightarrow 31.5 \rightarrow 1.8 \rightarrow 0.0$ mV/ns across 10/15/25/35 ns. Past 15 ns it only costs conduction loss. results/gan_analysis.m, results/gan_master.m — both run in MATLAB or Octave.

Result 4 — the two halves nobody separates

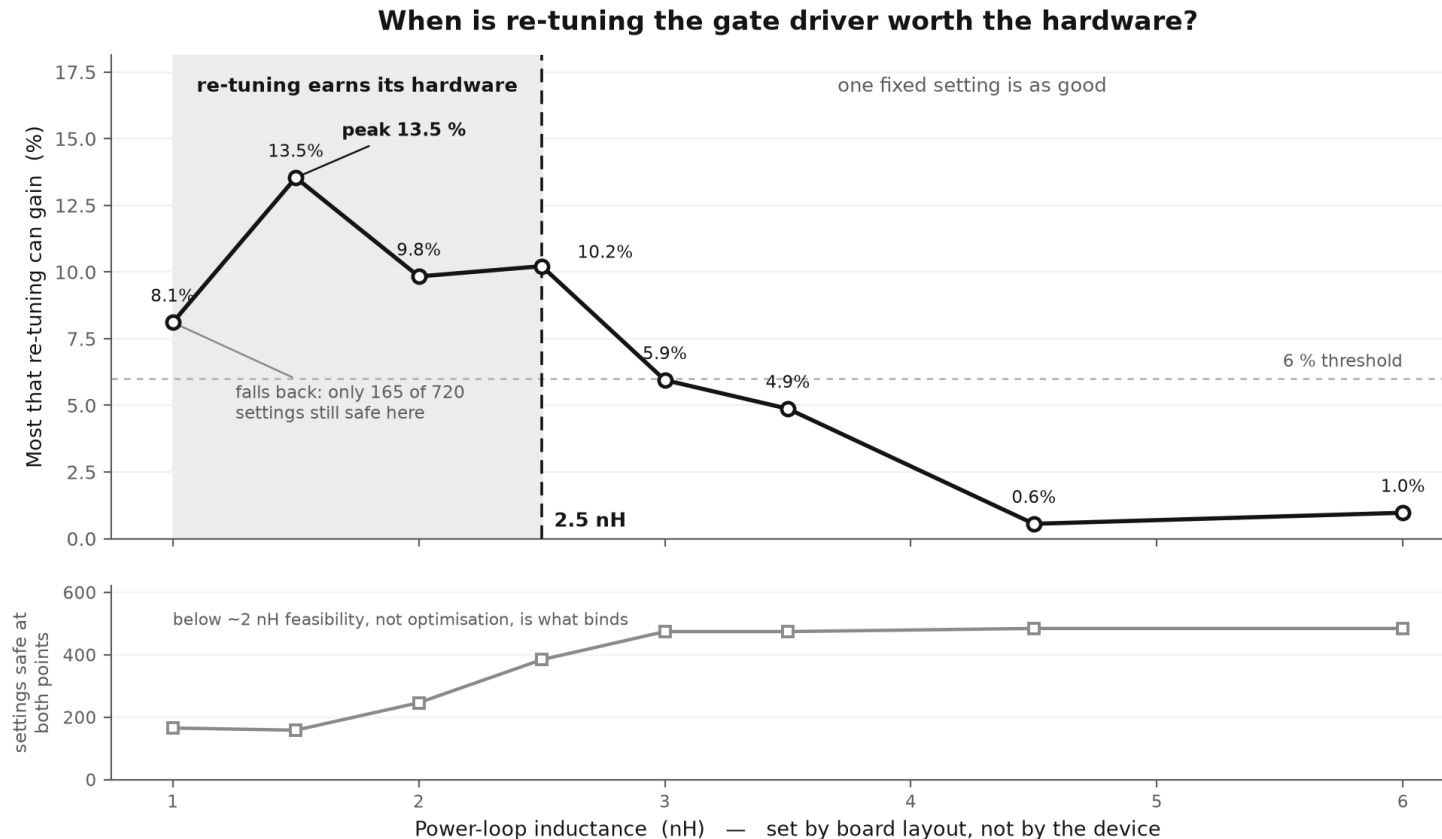
The base paper [9] (doi.org/10.1002/cta.3136) shows that a gate waveform can be chosen by a digital code. It, and every paper after it, then reports one number — how much better than a plain driver. That number adds together two separate effects, and only the second needs a sensor, an ADC and a lookup table. Separating them means running every setting at every operating point, which is why nobody has.

(A) Pick a better FIXED setting	25.1 %	no sensing · no LUT
(B) CHANGE it per operating point	3.9 %	needs all of it
(B') ...but ONE comparator gets 72 % of (B)	2.8 %	a threshold, not a LUT

So the full sensor + ADC + lookup table is left justifying 3.7 % of the total gain, over a fixed setting plus one comparator. Re-tuning is 13.4 % of the gain; a single threshold takes 72 % of that.

Every figure uses the same baseline — the middle setting that is safe at all four operating points. The split does not depend on the weighting: across 106 weightings, (A) stays 23.4–29.0 % and (B) 1.3–6.4 %, and (A) beats (B) every time. `scripts/novelty.py`, `scripts/weight_sensitivity.py`.

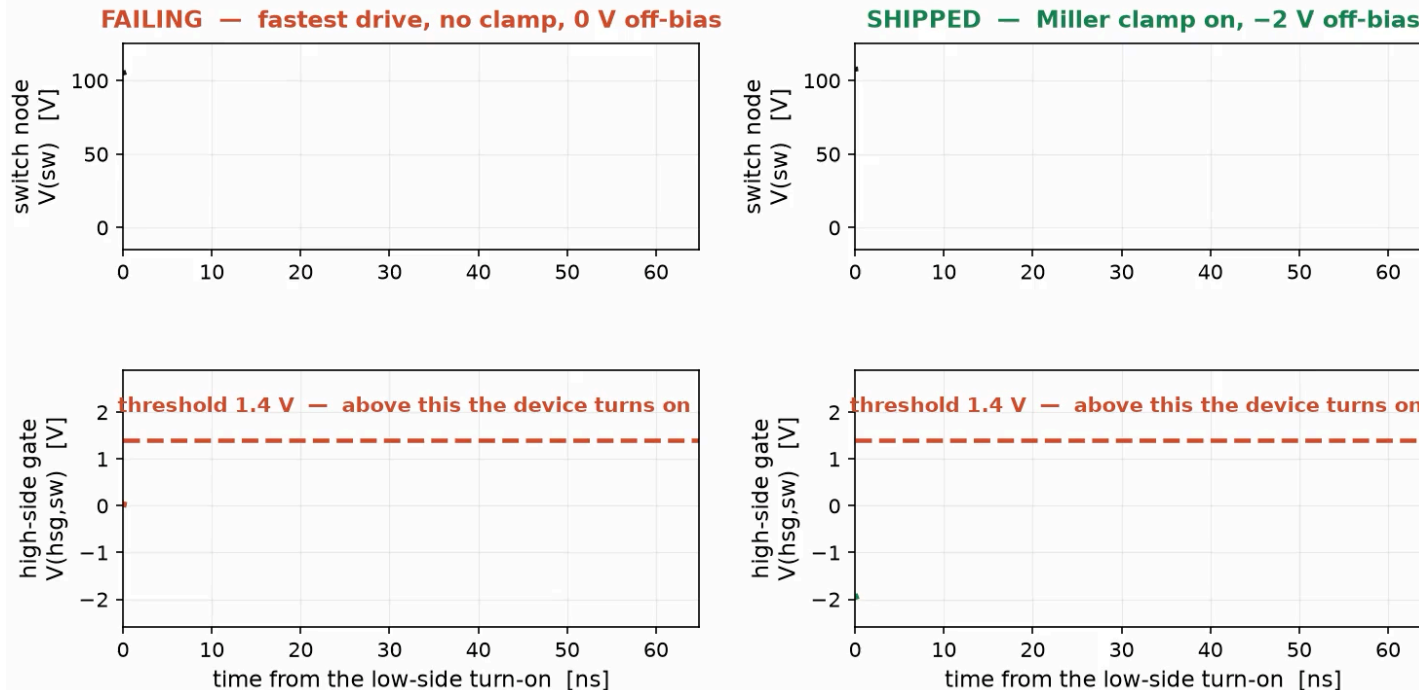
Result 5 — re-tuning only pays below ~2.5 nH



All 720 settings at two operating points for every value, 7,200 runs. Excluding clamp-chatter points changes nothing. scripts/loop_sweep.py, loop_analyse.py.

Eight inductance values, 7,200 runs. The answer is a band, not a single line: re-tuning only pays from about 2.5 nH downwards, peaking at 13.5 % at 1.5 nH, then falling back to 8.1 % at 1.0 nH because only 165 of the 720 settings are still safe there — below about 2 nH the limit is what is safe, not what is best. A designer measures their own board and reads the decision off this curve.

The low-side device turns on. Watch the switch node fall.



Real ngspice output from sim/dpt.cir — only the pacing is presentational.

What it shows

- **TOP row: the switch node.** The low-side device turns on and the voltage collapses from 100 V in a few nanoseconds. That fast drop is the cause.
- **BOTTOM row: the gate of the OFF device.** The fast drop pushes charge through CGD and lifts that gate.
- **LEFT, failing:** the gate reaches +1.65 V, above the 1.4 V line drawn on the plot. The device turns on when it must not.
- **RIGHT, shipped:** same circuit, same setting, plus the gate clamp and -2 V rail. Peak -1.18 V, a 2.58 V margin. Both panels are the same simulation; only the clamp and the off rail differ.

Click to play (22 s). Real ngspice waveforms from sim/dpt.cir, run twice — only the pacing and the captions are presentational.

Conclusion & next steps

What the data supports

- Picking the setting well: 25.1 %. Changing it per operating point: 3.9 %.
- One comparator gets 72 % of that 3.9 %.
- **What to build:** one fixed setting plus a light-load comparator. The full sensor + ADC + lookup table is left justifying 3.7 %.
- The FPGA half is real: 20 LUTs and 20 flip-flops on an xc7a35t, 200 MHz met with 1.996 ns to spare.

What is next

- Review-II — transistor-level output stage in Cadence; re-run the answer on real devices.
- Review-III — measure a hardware half-bridge. Until then this is a simulation study, and is titled as one.
- Limits we state ourselves: no silicon measured; one device model underlies everything.

References (1–15)

- [1] Z. Zhang, F. Wang, L. M. Tolbert and B. J. Blalock, "Active Gate Driver for Crosstalk Suppression of SiC Devices in a Phase-Leg Configuration," IEEE Trans. Power Electron., vol. 29, no. 4, pp. 1986–1997, Apr. 2014.
- [2] R. Xie, H. Wang, G. Tang, X. Yang and K. J. Chen, "An Analytical Model for False Turn-On Evaluation of High-Voltage Enhancement-Mode GaN Transistor in Bridge-Leg Configuration," IEEE Trans. Power Electron., vol. 32, no. 8, pp. 6416–6433, Aug. 2017.
- [3] D. Reusch and J. Strydom, "Understanding the Effect of PCB Layout on Circuit Performance in a High-Frequency Gallium-Nitride-Based Point of Load Converter," IEEE Trans. Power Electron., vol. 29, no. 4, pp. 2008–2015, Apr. 2014.
- [4] B. Li, G. Zhang, C. Li, G. Wang, S. Liu and D. Xu, "Crosstalk Suppression Method for GaN-Based Bridge Configuration Using Negative Voltage Self-Recovery Gate Drive," IEEE Trans. Power Electron., vol. 37, no. 4, pp. 4406–4418, 2022.
- [5] X. Chen, H. Peng, S. Song, Q. Tong and Y. Kang, "A Novel Control Strategy for Optimal Tradeoff between Overshoot and Switching Loss Based on Double Closed-Loop Self-Regulating Active Gate Driver," IEEE Trans. Power Electron., vol. 39, no. 10, pp. 13033–13043, 2024.
- [6] W. Song, T. Hu, J. Chen, T. Tang, H. Yue and G. Liu, "A Self-Regulating Active Gate Driver of Voltage Overshoot Suppression for SiC MOSFETs Under Variable Load Current Conditions," IEEE Trans. Power Electron., vol. 40, no. 8, pp. 10623–10634, 2025.
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